

SUMMARY

ML Engineer specializing in trustworthy and explainable machine learning, with hands-on experience building end-to-end ML solutions, from optimization and model development to interpretability and robust deployment. Strong research and engineering background with a focus on healthcare AI and real-world applications.

EDUCATION

Higher Institute Of Computer Science Tunis El Manar	
Research master in Software engineering - Ranked 1st in my cohort (Laureate)	2024 - July, 2026
Bachelor in Computer Science	2021 - 2024

RESEARCH & PROFESSIONAL EXPERIENCE

Research Master's End of Studies Internship – SERCOM Laboratory (EPT):	01/2026-07/07/2026
Designed a hybrid ML pipeline that combined unsupervised K-Means clustering with supervised learning to enrich feature representations and improve diabetes prediction, leveraging systematic hyperparameter optimization and XAI; co-authored a journal article submitted to <i>Biomedical Signal Processing and Control</i>	
Internship – SERCOM Laboratory (EPT): Transparent AI and Explainability	03/2025 - 08/2025
Developed and optimized interpretable ML models for healthcare prediction, applying data balancing, GridSearchCV, Bayesian Optimization, Optuna, and Genetic Algorithms; contributed to peer-reviewed research and an ICAART submission.	
Independent Research Project – Generative Artificial Intelligence: Conducted an independent review of Generative AI and LLMs, synthesizing advances, applications, and research challenges into a scientific paper accepted at ICCAD.	
End of Studies Project – DataHorizon: Developed a process management application integrating APIs and automated workflows using React, Angular, C#, SharePoint/SPFx, Azure Functions, and Microsoft Graph API.	
Summer Internship – SNDP (Agil Project): Developed an end-to-end online sales for lubricants platform using PHP, JavaScript, Bootstrap, and MySQL.	
Co-Founder – Digourdia (Startup Project): Co-developed an e-commerce MVP, contributing to UI/UX, prototyping, and business model development; won 2nd Prize at Hult Prize OnCampus.	

SKILLS

- **Programming Languages:** Python, R, SQL, Matlab, C#, C++, PHP
- **Database Management:** MySQL, PostgreSQL, Microsoft Graph API
- **Frameworks / Libraries:** TensorFlow, PyTorch, SciPy, Pandas, NumPy, Scikit-learn, OpenCV, YOLOv8, Bootstrap
- **Web Development:** HTML, CSS, JavaScript, ReactJS, NodeJS, Angular, SharePoint SPFx
- **Tools & Platforms:** Git/GitHub, LaTeX, Google scholar, Google Colab, Cisco Packet Tracer, Jupyter Notebook

CERTIFICATIONS & COURSES

- Fundamentals of Deep Learning NVIDIA
- Data Science Tools IBM
- Data Science Methodology IBM
- Build Your Own Chatbot IBM (onGoing)
- Generative AI with Diffusion Models NVIDIA (onGoing)

VOLUNTEERING & LEADERSHIP

- **International:** Tunisian student representative at Global Next Leaders Forum (Japan) | Co-Program Manager - YESIST12 Host Committee 2024 | Organizing Committee - IEEE R8SYP 2022 | Marketing Lead – IEEE KTM | IEEE Smart Village Team member
- **National:** Vice Chair - IEEE EMBS Tunisia | Coordinator - IEEE Tunisia SIGHT
- **University & Community:** Former Chair - IEEE ISI SB | Co-Founder & Project Coordinator - IEEE SIGHT ISI SBG Founder - ISI Events | Lead Organizer, ISI In Motion 1.0 & 2.0; Member of Securinets ISI and Freeways clubs

LANGUAGES

Arabic - Native/Bilingual | French - Proficient | English - Proficient

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